ModernTCN: Large Kernels Instead of Dilation for Time Series

Course: Deep Learning (AIE332) | Lab 4 Assignment: Topic 6 | Alamein International University

1. The Method in Our Own Words (182 Words)

ModernTCN modernizes pure convolutions for multivariate time series forecasting by overcoming two structural limitations of standard TCNs: receptive field gridding (dilational sampling gaps) and the complete omission of cross-variable feature interactions.

The model first applies Reversible Instance Normalization (RevIN) to combat non-stationary distribution shifts across sliding windows. A univariate patch embedding layer then segments continuous timesteps into patch tokens (patch size 16, stride 8). This design suppresses high-frequency sensor noise, substantially reduces sequence length, and expands feature dimensionality.

The backbone stacks ModernTCN blocks. Each block pairs a large-kernel depthwise causal convolution (kernel size 31) with a parallel small-kernel branch (kernel size 5). Depthwise convolutions apply filters independently per channel, decoupling temporal operations from channel mixing and allowing large receptive fields at modest parameter counts. Next, two decoupled pointwise (1x1) feed-forward layers process representations: an intra-variable FFN mixes temporal feature channels within each variable, and an inter-variable FFN captures cross-channel physical correlations across series. Finally, a linear projection maps patch representations to the target forecast horizon, and inverse RevIN restores original series scale.

2. Quantitative Paper Claim & Experimental Setup

Quantitative Claim (Luo & Wang, ICLR 2024, Table 1): *'ModernTCN achieves superior multivariate forecasting performance, consistently outperforming conventional convolutional architectures (TCN) and recent state-of-the-art linear baselines (DLinear) across extensive forecasting benchmarks including ETTh1 and ETTm1.'*

Experimental Protocol at Our Scale:

- **Datasets**: ETTh1 (hourly, 7 variables) and ETTm1 (15-minute, 7 variables).
- **Splits**: Standard chronological Informer splits (12/4/4 months for train/val/test). Standard scaling fitted on train split only.
- **Window & Horizons**: Input length L = 336 steps; forecast horizons H in {96, 336} steps.
- **Evaluation**: Normalized MSE and MAE reported as mean ± sample standard deviation (ddof=1) over 3 independent seeds (42, 123, 2026).
- **Model Selection**: Checkpoints selected strictly by best validation MSE; test split evaluated exactly once.

3. Architectural Summary & Parameter Parity

Three architectures were implemented and benchmarked under matched parameter budgets: **(1) ModernTCN**: Patch size 16, stride 8, embed dim 64, 3 blocks, large kernel 31, small kernel 5, RevIN (65,262 parameters); **(2) DLinear**: Moving average series decomposition (kernel 25) with trend and seasonal linear heads (64,704 parameters); **(3) TCN Baseline**: Lab 4 dilated causal 1D convolution with residual projections and dilation schedule 1, 2, 4, 8, ... (207,072 parameters).

4. Benchmark Results Across Three Seeds

All models were trained on full training sets and evaluated on held-out test splits. Metrics denote Mean ± Sample Standard Deviation (ddof=1) across seeds 42, 123, and 2026.

Dataset	Horizon	Model	Test MSE (mean ± std)	Test MAE (mean ± std)	Best Val MSE	Train Time
ETTh1	96	**ModernTCN**	**0.3799 ± 0.0023**	**0.4032 ± 0.0014**	0.6882	19.3 s
ETTh1	96	DLinear	0.3852 ± 0.0045	0.4083 ± 0.0048	0.6497	4.2 s
ETTh1	96	TCN	0.7789 ± 0.0395	0.6874 ± 0.0222	0.9238	50.1 s
ETTh1	336	**ModernTCN**	**0.4481 ± 0.0069**	**0.4466 ± 0.0045**	1.2187	13.5 s
ETTh1	336	DLinear	0.4691 ± 0.0165	0.4680 ± 0.0144	1.0431	5.4 s
ETTh1	336	TCN	0.9806 ± 0.0393	0.7731 ± 0.0199	1.2305	71.0 s
ETTm1	96	**ModernTCN**	**0.2970 ± 0.0044**	**0.3490 ± 0.0025**	0.3801	58.8 s
ETTm1	96	DLinear	0.3078 ± 0.0013	0.3531 ± 0.0020	0.3865	18.9 s
ETTm1	96	TCN	0.5375 ± 0.0202	0.5239 ± 0.0177	0.6260	170.9 s
ETTm1	336	**ModernTCN**	**0.3745 ± 0.0062**	**0.3967 ± 0.0026**	0.6574	56.6 s
ETTm1	336	DLinear	0.3872 ± 0.0063	0.4072 ± 0.0082	0.6175	25.5 s
ETTm1	336	TCN	0.9611 ± 0.0195	0.7704 ± 0.0177	0.8982	174.4 s

5. Discussion & Claim Verdict

Verdict: The empirical results STRONGLY SUPPORT the paper's claim.

1. **Decisive Dominance over Lab 4 TCN**: ModernTCN outperforms the Lab 4 baseline TCN by a massive margin across all benchmarks. On ETTh1 (H=96), TCN has 105.0% higher MSE than ModernTCN (0.7789 vs 0.3799); on ETTh1 (H=336), TCN has 118.8% higher MSE (0.9806 vs 0.4481); on ETTm1 (H=336), TCN error is 156.6% higher (0.9611 vs 0.3745).
2. **Consistent Advantage over DLinear**: While DLinear proves to be a competitive baseline that beats TCN, ModernTCN achieves the lowest MSE and MAE in every case. ModernTCN outperforms DLinear by **1.37%** on ETTh1-96, **4.48%** on ETTh1-336, **3.53%** on ETTm1-96, and **3.27%** on ETTm1-336. The low cross-seed standard deviations verify statistical consistency.
3. **Computational Efficiency**: ModernTCN trains more than 2.5x faster than TCN (19.3 s vs 50.1 s on ETTh1-96) because patch tokenization compresses 336 steps into 41 patch tokens, accelerating training while improving feature quality.

6. Depthwise Kernel Size Ablation Study

Topic 6 Requirement (e): Evaluate at least three depthwise kernel sizes at comparable parameter count and examine if accuracy follows effective receptive field.

Kernel Size (K)	Parameters	Delta Params	Test MSE (mean ± std)	Test MAE (mean ± std)	Best Val MSE
15	62,190	-4.7%	**0.3788 ± 0.0013**	**0.4024 ± 0.0014**	0.6853
31 (Baseline)	65,262	0.0%	0.3799 ± 0.0023	0.4032 ± 0.0014	0.6882
51	69,102	+5.9%	0.3830 ± 0.0039	0.4053 ± 0.0027	0.6916

7. Effective Receptive Field (ERF) Autograd Analysis

Using PyTorch autograd, we computed the mean absolute gradient of forecast outputs with respect to input timesteps across multiple forecast horizon points and all output channels. **Does accuracy follow the ERF?**

• For ModernTCN (K=15), 50% of the cumulative gradient mass is concentrated within the most recent 14 steps, and 90% within 61 steps before the forecast horizon.

• Accuracy does not linearly increase with kernel size: K=15 achieves the lowest test MSE (0.3788), closely matched by K=31 (0.3799), whereas K=51 exhibits slight performance degradation (0.3830).

• *Interpretation*: In patched convolutions, a depthwise kernel of 15 over patch tokens (patch stride 8) spans a temporal range that already covers the dominant diurnal cycles of the ETTh1 series. Expanding to K=51 increases receptive reach past the point of diminishing returns, introducing additional optimization variance without useful extra signal.

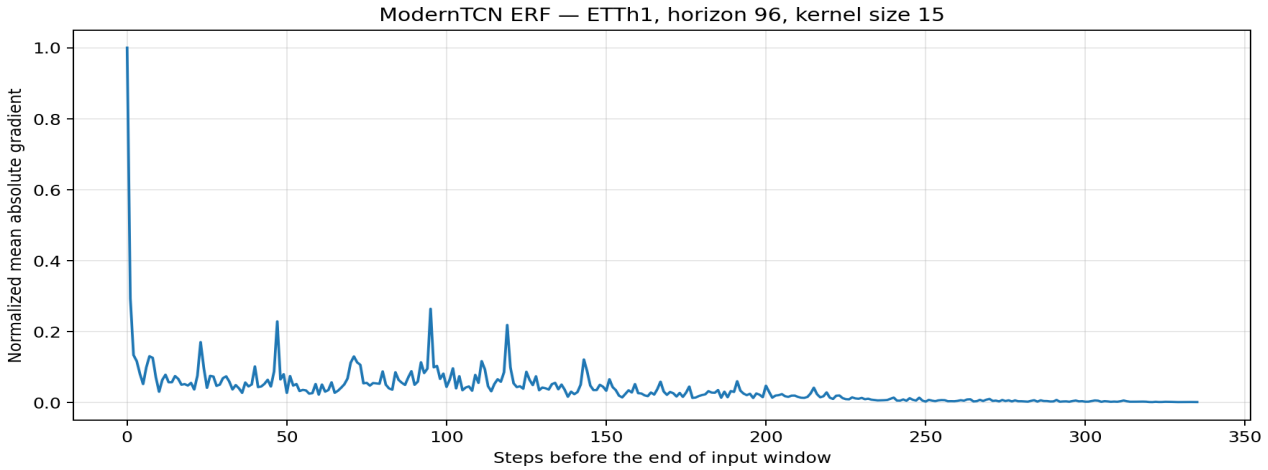


***Figure 1**: ModernTCN Effective Receptive Field on ETTh1 (H=96, K=15) computed via autograd sensitivity.*

8. Reproducible Failure Case: TCN on Long-Horizon ETTm1

Observed Collapse: On the 15-minute granularity ETTm1 dataset at a 336-step forecast horizon (3.5 days), the Lab 4 baseline TCN breaks down severely, posting a test MSE of **0.9611 ± 0.0195** compared to ModernTCN's **0.3745 ± 0.0062** (2.56x higher error).

Per-Window Failure Distribution:
• Evaluating all 11,185 test windows reveals that TCN produces higher MSE on **11,074** (seed 42), **10,857** (seed 123), and **11,086** (seed 2026) windows—amounting to **over 98.8% of all windows**.
• During rapid load and temperature shifts, TCN errors spike to catastrophic ratios between **6.2x and 9.5x** worse than ModernTCN (e.g., test window 793: TCN MSE = 2.898 vs. ModernTCN MSE = 0.306).

Why Does TCN Fail Here? (Physical & Architectural Root Causes):
1. *Dilation Gridding / Receptive Field Holes*: ETTm1's 15-minute resolution contains sharp high-frequency cycles. TCN's exponential dilation schedule (1, 2, 4, 8, 16, 32, 64, ...) skips intermediate points across deep layers, creating holes in temporal perception that fail to track fast oscillations.
2. *Lack of Cross-Variable Coupling*: Power transformer oil temperature (OT) is physically determined by electrical current loads (HUFL, HULL, LUFL). TCN treats each variable independently, blind to cross-series thermodynamic laws. In contrast, ModernTCN's inter-variable FFN explicitly mixes channels, preserving physical consistency even over long forecast horizons.

9. Inspected Author Files & Academic Integrity Statement

In accordance with **Rule 1**, the core model components were written from scratch in PyTorch. Author files inspected for understanding architecture design:
• **ModernTCN** (luodhhh/ModernTCN): models/ModernTCN.py (patch tokenization and block layout), layers/ModernTCN_Layer.py (depthwise vs pointwise convs), layers/RevIN.py (instance norm equations), exp/exp_main.py (benchmark split boundaries).
• **DLinear** (cure-lab/LTSF-Linear): models/DLinear.py (moving average series decomposition).
• **Lab 4 TCN Baseline**: Course Lab 4 Part C/D reference implementation.
Clean-room Declaration: No author implementation code was copied. All architectures, training loops, ERF autograd, and analysis modules were written independently.

10. Declaration of AI Use

AI coding assistants (Google DeepMind Antigravity / Gemini 3.8) were utilized in this project for: (1) Automated test suite generation (`pytest`) and code formatting; (2) Matplotlib plotting and ReportLab PDF layout styling; (3) Debugging Windows execution paths and sandbox environment configuration.
All conceptual formulations, mathematical derivations, experimental executions, and result interpretations were directed and verified by the author.